

Stage (i)

Coarse seeding

- local search near expected wavelength
- peaks ranked by proximity and depth
- offset guard and close-pair split

Stage (ii)

Adaptive windowing / morphology checks

- per-line window within min/max bounds
- expansion or contraction
- morphology checks trigger safer handling

Stage (iii)

Iterative bounded fitting

- Voigt / asymmetric-Voigt fit
- recenter and iterate to convergence
- fallback refit if needed; output centre, velocity, intensity